



ToolShare

University Equipment Sharing Platform

SSE Capstone — Project Day

Our Team



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The Problem

Problem

- No centralized system for equipment management
- Manual tracking wastes staff time
- Students struggle to find available tools
- Equipment gets lost or damaged
- Little accountability or tracking history

The Solution

Solution

- Centralized equipment management system
- Easy booking & reservation workflow
- Role-based access (Students / Faculty / Admin)
- Automated approvals and notifications
- Condition tracking with image uploads

Who Uses ToolShare?



Student

Browses and requests lab equipment for academic use.



Faculty / Lab Instructor

Approves or rejects requests, handles checkout & returns.



Admin

Full system access: manages users, items, RBAC, and analytics.

Why ToolShare?



No Visibility

No way to know if an item is available or already borrowed.



Manual Tracking

Staff manage requests via email and paper — error-prone and slow.



Lost Equipment

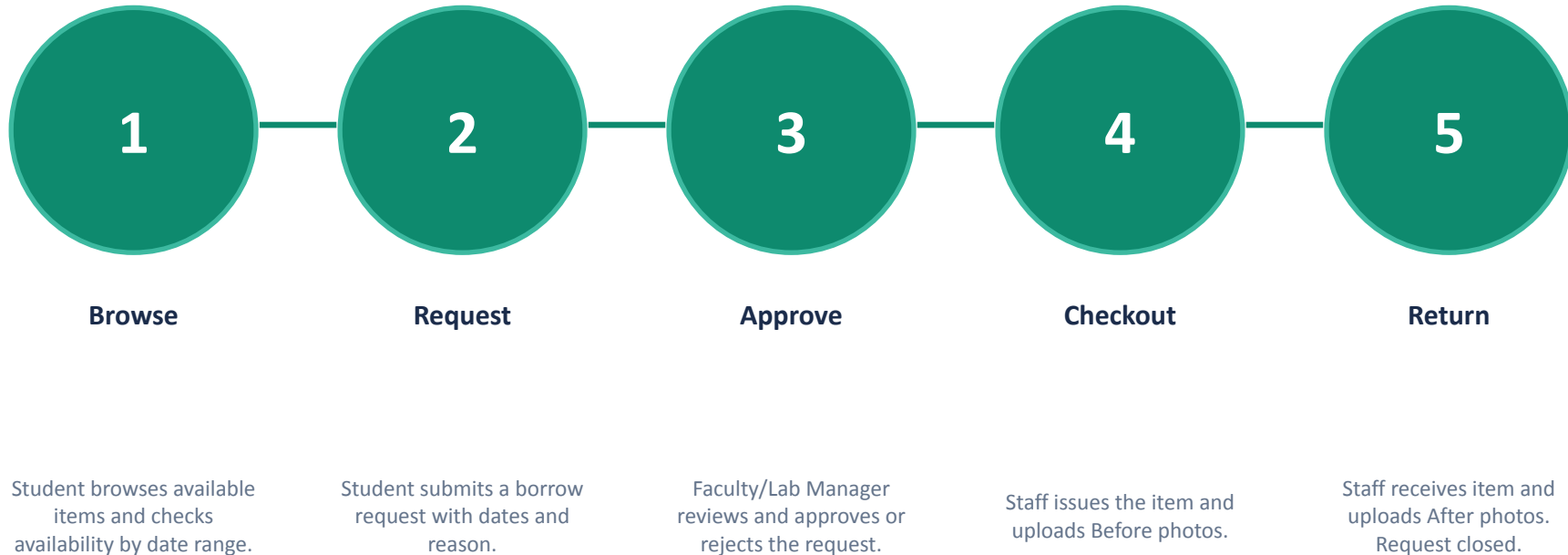
No condition photos means no accountability when items are returned damaged.



No Reminders

Overdue returns go unnoticed with no automated follow-up system.

How It Works



Key Features



Availability Filter

Real-time item search filtered by date range.



Role Based Access Control

Role-based permissions for Student, Staff, and Admin.



Condition Photos

Before & After images uploaded at checkout and return.



Notifications

In-app and email alerts at every workflow stage.



Overdue Detection

Scheduled jobs flag overdue returns automatically.



Analytics

Usage metrics, overdue and borrowing trends.

What Makes Us Different?

Feature	Email/Paper	Generic Booking	ToolShare
Online Requests	✗	✓	✓
Condition Photos	✗	✗	✓
RBAC Permissions	✗	✗	✓
Auto Overdue Alerts	✗	✗	✓
Admin Dashboard	✗	Partial	✓

Tech Stack

Frontend

React (Vite)



Backend

Node.js + Express



Database

MySQL



Deployment

aws Cloud



Notifications

Email and In app



Scheduler

Email Notification



In a lab environment, time is a resource. ToolShare makes sure none of it is wasted on paperwork.



Problem Solved

Replaces fragmented email/paper workflows with a fully digital, accountable borrow lifecycle.



Built for Universities

Designed for students, faculty and lab managers with role-based access at every step.



Data-Driven Oversight

Analytics, audit logs, and automated overdue detection give administrators full visibility.



DEMO

Future Work

1

Mobile App

Native iOS/Android app for on-the-go borrowing and push notifications.

2

Barcode Scanning

Scan item barcodes or QR codes at checkout and return for faster processing.

3

University SSO Integration

Login using university accounts

4

Fine-grained Analytics

Per-lab, per-department usage reports and exportable CSV/PDF summaries.

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Thank You! — Any Questions?